

ReReReRe: A Permutation-Based Coherence Detector

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Chapter 3 of the thesis

Motivation and Core Idea

Chapter 1 closed on a gap. The detection methods in widest use are strongest on *consistent* careless responding—straightlining, acquiescence, machine-like regularity—and weakest on *inconsistent* careless responding, where a respondent’s answers vary across items but bear no coherent relationship to item content. Random and mixed responding are the hardest patterns to prevent and the most damaging to covariance-based analyses, yet the only established index that targets them directly, the psychometric-synonym statistic (Meade & Craig, 2012), operates on a handful of hand-selected item pairs. This chapter introduces *ReReReRe*, a detector built specifically to exploit the full correlational structure of a multi-construct questionnaire in order to identify inconsistent careless responding.

The core idea is a single observation made operational. An attentive respondent’s answer to one item is informative about their answer to a correlated item, because both items load on a shared latent factor (Section 2.4). A careless respondent’s answers to the same pair are *incoherent*: the cross-item structure that the sample as a whole exhibits is absent in their individual profile. ReReReRe measures, for each respondent, how much of the sample’s pairwise structure their own answers reproduce, and compares that quantity against a baseline obtained by permutation. The result is a per-respondent z-score: high when the respondent is more coherent than chance (attentive), low or negative when they are no more coherent than randomly paired items would predict (potentially careless). Because the baseline is built separately for each respondent, the method is *self-calibrating*: it makes no distributional assumption about the coherence statistic, sidestepping the normality problems that, as Chapter 1 noted, cripple the chi-square cutoff for Mahalanobis distance on Likert data.

The Algorithm

Individual-level coherence

Let X be the $N \times J$ matrix of responses (N respondents, J items). The sample-level correlation matrix R summarizes how items relate across respondents. ReReReRe selects a set of k item pairs $\{(a_1, b_1), \dots, (a_k, b_k)\}$ that the sample deems related (Section 3.2.2), and then, for each respondent r , forms the two vectors of that respondent’s answers to the left-hand and right-hand members of the pairs,

$$\mathbf{u}_r = (X_{r,a_1}, X_{r,a_2}, \dots, X_{r,a_k}), \quad \mathbf{v}_r = (X_{r,b_1}, X_{r,b_2}, \dots, X_{r,b_k}).$$

The respondent’s *coherence* is a weighted association between $\mathbf{u}_r$ and $\mathbf{v}_r$, with each pair contributing in proportion to its sample-level $|r|$,

$$C_{\text{obs}}(r) = \sum_{j=1}^k w_j \tilde{u}_{r,j} \tilde{v}_{r,j}, \quad w_j = \frac{|R_{a_j b_j}|}{\sum_l |R_{a_l b_l}|}, \quad (1)$$

where $\tilde{u}, \tilde{v}$ denote the item-standardized values. Strong pairs count more than weak ones, but no pair is discarded outright. The crucial point is that Equation 1 is computed *down a respondent’s row*, not across respondents: it asks whether *this person* answered the coupled pairs coherently.

Pair selection: two regimes

How the k pairs are chosen depends on questionnaire length, because the signal ReReReRe relies on comes from the total number of high-correlation pairs available, and that number scales very differently for short and long instruments. ReReReRe therefore switches automatically between two regimes at a threshold of 60 items (the `mode = "auto"` default).

Weighted regime (used for $J \leq 60$ items). Every one of the $\binom{J}{2}$ item pairs is included, each weighted by its absolute sample correlation as in Equation 1. With short questionnaires there are too few strongly correlated pairs to select a clean subset, so discarding any of them throws away signal; using all pairs but down-weighting the weak ones extracts the most from limited structure.

Coupled regime (used for $J > 60$ items). Only the top `corProp` fraction of pairs by absolute correlation is retained (default 3%), subject to a floor of `min_pairs = 15`. With long questionnaires the strongest pairs carry a clean, redundant signal, and restricting attention to them sharpens the contrast between attentive and careless respondents while keeping the computation tractable.

The switch is automatic and recorded in the output (`mode_used`); a user can override it to force either regime. Calibration studies confirm a crossover near 60 items: below it the weighted regime wins by up to 0.06 MCC, above it the coupled regime wins by a similar margin.

The permutation baseline and the z-score

A high coherence is only meaningful relative to what coherence the same respondent would show over pairs with *no* shared structure. ReReReRe constructs this baseline by permutation (Section 2.6). For each of `iterations = 100` rounds, it draws k *random* item pairs from the matrix, recomputes the same weighted coherence as in Equation 1, and thereby obtains, for each respondent, a null distribution $\{C_{\text{rand}}(r, 1), \dots, C_{\text{rand}}(r, 100)\}$. The respondent’s score is the standardized distance of their observed coherence from this personal baseline,

$$z_{\text{RR}}(r) = \frac{C_{\text{obs}}(r) - \text{mean}_s C_{\text{rand}}(r, s)}{\text{sd}_s C_{\text{rand}}(r, s)}. \quad (2)$$

This is the permutation z-score of Equation 2.3, specialized to coherence. A respondent is flagged as careless when z_{RR} falls below a threshold (Section 3.5). The algorithm is summarized in the box below.

Table 1*Default parameters of ReReReRe() and their role.*

Parameter	Default	Role
mode	"auto"	Weighted if $J \leq 60$, coupled if $J > 60$
corProp	0.03	Fraction of top- $ r $ pairs kept (coupled regime)
min_pairs	15	Floor on the number of coupled pairs
iterations	100	Number of permutation rounds for the baseline
align_signs	TRUE	Reverse-code negatively-correlated pairs
rescale	"proportion"	Per-item rescaling for mixed scales
z_threshold	1.5	Flagging cutoff (or <code>auto</code>)
auto_z	FALSE	Calibrate the cutoff from total item count

Algorithm: ReReReRe (standard mode).

1. Rescale each item (default: divide by its observed maximum; Section 3.4).
2. Compute the sample correlation matrix R ; record the sign of each correlation.
3. *Sign alignment*: for each pair with $R_{ab} < 0$, reverse-code column b (Section 3.3).
4. Select pairs: all pairs (weighted, $J \leq 60$) or the top- $k\%$ by $|r|$ (coupled, $J > 60$).
5. Compute each respondent's weighted coherence C_{obs} (Eq. 1).
6. Repeat 100 times: draw k random pairs, sign-align, compute C_{rand} .
7. Convert to a per-respondent z-score z_{RR} (Eq. 2).
8. Flag respondents with $z_{\text{RR}} \leq \text{threshold}$ (default 1.5, or auto-calibrated).

In R the entire procedure is a single call, `ReReReRe(data)`, returning a data frame with each respondent's z_{RR} , the flag, and metadata (the regime used, the calibrated threshold, the estimated number of factors). The default parameters are listed in Table 1.

Reverse-Coded Items: Sign Alignment

Questionnaires routinely include *reverse-keyed* items, worded so that agreement indicates *less* of the trait. Such an item correlates *negatively* with same-trait items keyed in the usual direction. This creates a subtle but fatal problem for any method that aggregates correlations across pairs. If the coherence of Equation 1 mixes positively-keyed pairs (which contribute positive products for an attentive respondent) with negatively-keyed pairs (which contribute negative products), the two cancel inside the sum, and even a perfectly coherent respondent receives $C_{\text{obs}} \approx 0$. Every respondent then looks equally incoherent, and the z-score carries no signal.

The defect is not hypothetical. It was discovered on the Robie HEXACO-60 data, where 52% of the coupled pairs were oppositely keyed: with the pairs left unaligned, every respondent received $z_{\text{RR}} \approx 0$ (mean observed coherence 0.078 versus a baseline mean of 0.084), and the detector's area under the curve sat at chance. The fix, controlled by `align_signs` (default TRUE), is to use the *sign* of each pair's sample-level correlation to align the pair before computing coherence: wherever $R_{ab} < 0$, column b is reverse-coded (mapped by $x \mapsto (\max_b + \min_b) - x$) so that the pair contributes positively for a coherent

respondent. The same alignment is applied to the random pairs of the permutation baseline, so the null distribution is comparable. With the fix in place on the Robie data, mean coherence rose to 0.299, the mean z-score to 3.49, and the area under the curve from 0.47 to 0.66.

Sign alignment is more than data hygiene: it is what lets ReReReRe operate on real questionnaires *without* requiring the user to recode reverse-keyed items by hand. The procedure is transparent and emits a warning only when it is disabled in the presence of negatively-keyed pairs.

Mixed Response Scales: Rescaling

A second real-data complication is *heterogeneous response scales*: a battery that mixes, say, 1–4 and 1–7 items. Because the Pearson correlation in Equation 1 is sensitive to the numerical spacing of the response options, longer-range items would otherwise dominate a respondent’s coherence simply by virtue of their larger spread. ReReReRe addresses this with an optional per-item rescaling step (**rescale**), applied before any correlation is computed.

The default, **"proportion"**, divides each item by its observed maximum. This choice is principled. Section 2.2 noted that the Pearson correlation is invariant to identical affine transforms applied to all items: on a *homogeneous*-scale questionnaire every column is divided by the same constant, so the coherence is provably unchanged—the rescaling costs nothing. On a *heterogeneous*-scale questionnaire it equalizes the contribution of items measured on different ranges. On a stress test of 80 items split between 1–4 and 1–7 scales, proportion rescaling lifted MCC from 0.32 (no rescaling) to 0.43, while leaving uniform-scale results numerically identical. More aggressive options are exposed for advanced users—**"minmax"** and **"zscore"**—and **"none"** reproduces the legacy behaviour while warning when item ranges are heterogeneous. The recommendation is nonetheless to keep response scales uniform where possible; rescaling is a safeguard, not a substitute for clean design.

Threshold Selection and Auto-Calibration

The z-score of Equation 2 is continuous; flagging requires a cutoff. Two findings make this less fraught than it sounds. First, the method is *not* sharply threshold-sensitive: across a wide range of conditions, any cutoff between roughly 1.0 and 2.0 performs near-optimally, with a flat plateau of detection quality. The recommended universal default is $z_{\text{threshold}} = 1.5$, interpreted as “the respondent’s coherence is 1.5 standard deviations above their own random baseline.”

Second, where one wants to do better than a universal default, the optimal cutoff has a clean and interpretable dependence on questionnaire size. Figure 1 shows the optimal threshold as a function of the number of factors, holding items per factor fixed. The curve is *U-shaped*: at very low factor counts the z-scores of attentive and careless respondents overlap heavily, forcing a low threshold to catch anyone; as the number of factors grows the two distributions separate, and a *higher* threshold becomes optimal because it buys specificity without sacrificing sensitivity. The blue point cloud also encodes detection quality through marker size, which grows visibly from left (small, $\text{MCC} < 0.15$) to right (large, $\text{MCC} > 0.5$): more factors mean both a higher optimal threshold and better detection.

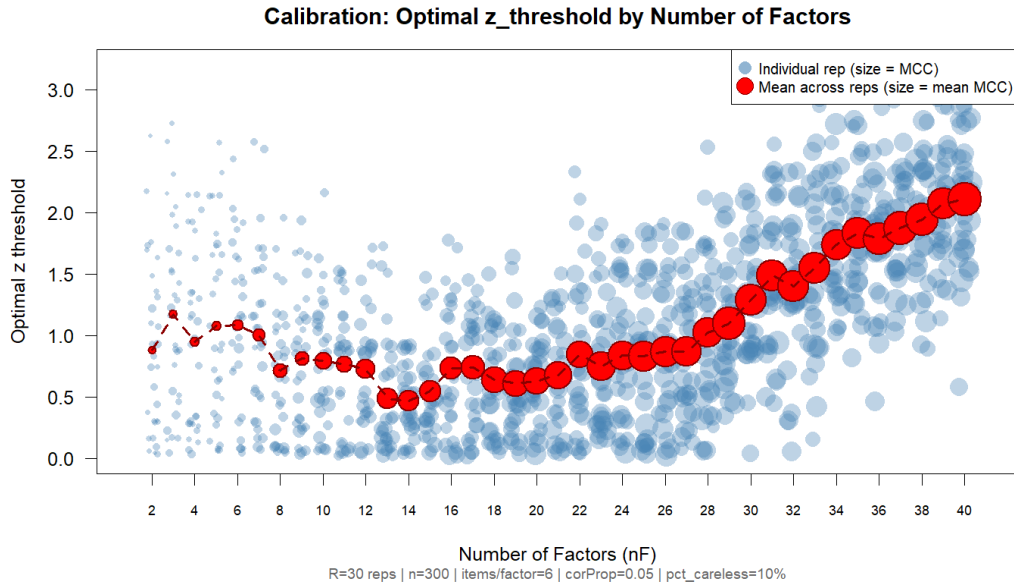


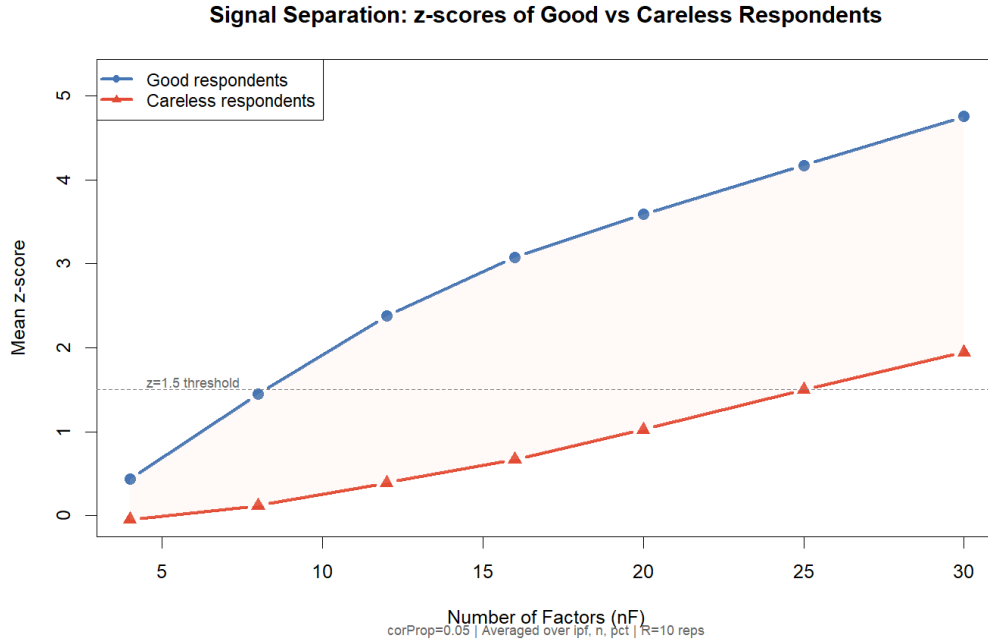
Figure 1

Optimal z -threshold as a function of the number of factors (items per factor fixed at 6, $n = 300$, 10% careless, 30 replications). Blue points are individual replications (size $\propto$ MCC); red points are replication means. The optimal threshold follows a U-shape, reaching a minimum near $z \approx 0.5$ at 13–15 factors and rising to $z \approx 2.1$ at 40 factors.

Rather than calibrate on the number of factors alone, the implementation uses the better predictor identified by a two-dimensional calibration study: the *total* number of items. A simple linear rule, $z \approx 0.505 + 0.0042 \times (\text{total items})$, explains roughly twice the variance of a rule based on the factor count, because what governs the separation of the z -score distributions is the total number of high-correlation pairs available, which scales with both the number of factors and the items per factor. Setting `auto_z = TRUE` activates this calibration: ReReReRe estimates the number of factors by parallel analysis (Section 2.4), looks up the calibrated threshold, and applies it. In aggregate, auto-calibration performs about as well as the fixed $z = 1.5$ default, confirming that the method does not demand careful tuning; its value is convenience, removing a parameter choice for non-technical users.

Intrinsic Properties

Before comparing ReReReRe to other detectors, it is worth characterizing how its performance depends on questionnaire structure, because those dependencies explain both its strengths and its limits. The picture below comes from a large simulation multiverse crossing the number of factors, items per factor, sample size, and careless base rate, with the structure of each dataset known by construction.

**Figure 2**

Mean z-score of attentive (“good”) versus careless respondents as a function of the number of factors (averaged over items per factor, sample size, and base rate; 10 replications). The separation grows quasi-linearly with dimensionality, explaining the method’s scaling advantage. The dashed line marks the $z = 1.5$ default cutoff.

Signal grows with dimensionality

The mechanism behind ReReReRe predicts that more factors should yield a cleaner separation between attentive and careless respondents, because a richer factor structure offers more coherent pairs against which incoherence stands out. Figure 2 confirms this directly. It plots the mean z-score of attentive and of careless respondents as the number of factors increases. At four factors the two groups overlap almost completely (a gap of about 0.48); by thirty factors they are separated by nearly three standard deviations (a gap of about 2.81). The attentive curve climbs steeply while the careless curve stays low, and the $z = 1.5$ default sits cleanly between them across most of the range.

Total length is the unifying predictor

Decomposing the variance in detection quality, the two strongest drivers are the items per factor ($\eta^2 = 0.31$) and the number of factors ($\eta^2 = 0.26$); sample size and base rate matter much less. Crucially, neither factor count nor items per factor alone is the right summary—their *product*, the total questionnaire length, is. Figure 3 plots detection quality against total items and collapses the separate effects onto a single saturating curve: MCC rises steeply up to roughly 150 items and then plateaus, with a slight decline beyond about 250 items where the number of items begins to approach the number of respondents and

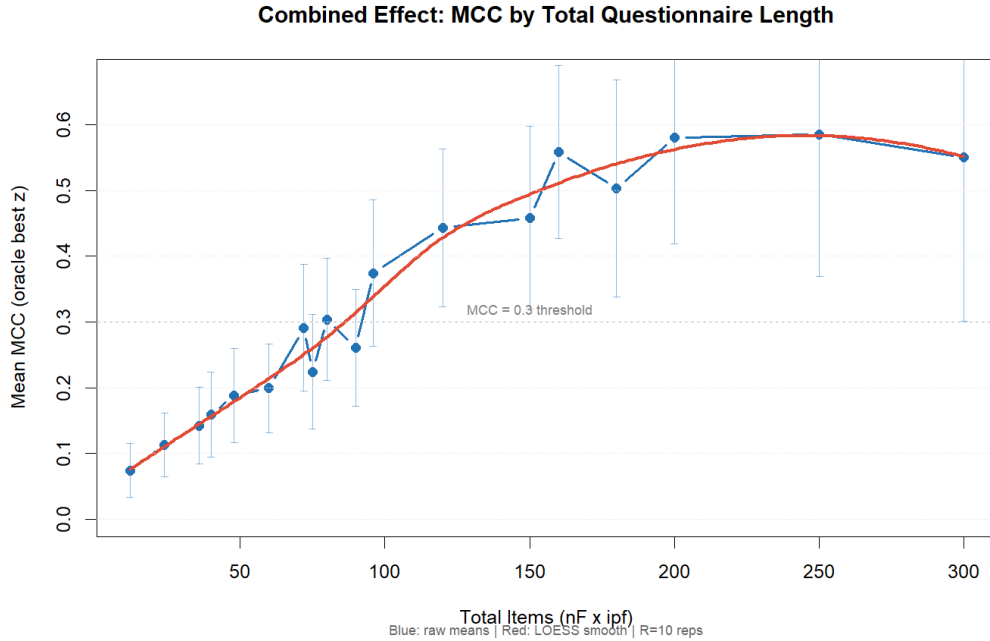


Figure 3

Detection quality (oracle-best MCC) as a function of total questionnaire length (number of factors $\times$ items per factor). Blue points are condition means with ± 1 SD bars; the red curve is a LOESS smooth. MCC saturates above roughly 150 items.

individual-level correlations grow noisy. The practical reading is that ReReReRe should be described in terms of *total length*, not number of constructs: a 120-item questionnaire of 12 factors with 10 items each detects far better than a 75-item questionnaire of 25 factors with 3 items each, despite the latter's higher factor count.

Detection by careless pattern

Broken down by the type of injected careless responding, ReReReRe detects all three inconsistent patterns at comparable rates: longstring at about 83%, mixed at about 81%, and random at about 79% (averaged over the favorable threshold range). This even profile is expected, since all three corrupt the cross-pair structure the method monitors. The limits of the method, examined next, lie not in any one inconsistent pattern but in the *consistent* patterns it is not designed to see.

Benchmark Comparison: ReReReRe versus Mahalanobis

Mahalanobis distance is the most widely used multivariate-outlier index for careless detection (Section 1.4.2), and it is the natural benchmark for ReReReRe. The comparison below was run on *identical* simulated datasets, so the two detectors face exactly the same data. Two framings are necessary, because the comparison hinges on how each method's threshold is set.

Practical thresholds: the decisive contrast

In practice, Mahalanobis distance is applied with a fixed cutoff: respondents whose squared distance exceeds the chi-square critical value at $\alpha = .001$, with degrees of freedom equal to the number of items, are flagged (Curran, 2016; Meade & Craig, 2012). This is the Mahalanobis counterpart of ReReReRe’s $z = 1.5$ default—a rule requiring no access to labels. Under this realistic, label-free comparison the two methods are not close. The chi-square rule is *essentially non-functional* on Likert data: its sensitivity is between 0.4% and 1.9%, so it catches almost no careless respondents, for an MCC of about 0.06. ReReReRe with its fixed default reaches an MCC of about 0.27 over the same conditions—more than four times higher. The reason is exactly the one developed in Chapter 1: the chi-square reference distribution assumes multivariate normality, which Likert data violate badly, so the nominal cutoff is far too conservative. ReReReRe’s permutation baseline makes no such assumption.

Oracle thresholds and the threshold-free view

It might be objected that Mahalanobis distance could do better with a tuned cutoff. This is true, but the tuning requires the true labels—an *oracle* no analyst possesses. Even granting Mahalanobis its oracle-best threshold per condition, ReReReRe (also at its oracle best) overtakes it as dimensionality grows: the two are tied near ten factors (MCC ≈ 0.37 for both), after which ReReReRe pulls ahead, reaching MCC ≈ 0.57 at twenty-five factors versus ≈ 0.39 for Mahalanobis, whose performance stays roughly flat. The fairest single comparison is the threshold-free area under the curve, which removes the thresholding question entirely (Section 2.10). Here ReReReRe’s AUC, computed with fully fixed defaults, climbs from 0.59 at four factors to 0.84 at twenty-five, overtaking Mahalanobis (flat at about 0.64) from eight factors upward. The conclusion is consistent across framings: Mahalanobis is a slightly better detector for very short questionnaires but does not scale, whereas ReReReRe scales strongly with questionnaire length and, under any realistic thresholding, dominates by a wide margin.

The two methods detect different things, and this is the deeper point. Mahalanobis distance flags multivariate *outliers*—respondents far from the sample centroid—whether careless or merely unusual. ReReReRe flags *incoherence*—a breakdown of the cross-factor structure. The next chapter exploits this complementarity rather than choosing between the two.

Design Alternatives Explored

The canonical ReReReRe() described above—the weighted/coupled auto-switch with sign alignment and proportion rescaling—was selected after a broad search over alternative formulations. They are noted here briefly for completeness; none displaced the canonical method as the recommended detector.

Factor-guided pair selection (an EFA-based variant, exposed as ReReReRe_F()) chooses within-factor pairs from an exploratory factor solution rather than by raw correlation. It wins on *simulated* data, where the factor structure is clean, but loses on *real* validation datasets, where parallel analysis on noisy, cross-loaded data produces poor factor assignments; the empirical top- $|r|$ ranking of the coupled regime proved more robust

to contamination. *Iterative* re-estimation (re-running the factor solution on the respondents judged attentive, then re-scoring) gave a small improvement on mid-length questionnaires but not elsewhere. A *split-half* scorer and a *variance penalty* (which pushes near-zero-variance straightliners toward negative z) each helped on specific partial-corruption or consistent-careless cases but did not improve overall classification enough to justify added complexity. The lesson from these experiments—that a method’s edge on simulated data need not survive on real data—shaped the validation philosophy of the chapters that follow. The variant code remains in the software for reference but is not part of the recommended pipeline.

Limitations and the Case for an Ensemble

ReReReRe is a strong standalone detector of inconsistent careless responding, but it has three limits, and naming them sets up the next chapter. First, it is *weak at low dimensionality*: with few factors the z -score distributions of attentive and careless respondents overlap (Figure 2), and detection is modest below roughly eight factors. Second, by construction it targets *inconsistent* careless responding and adds little on *consistent* patterns—pure straightlining and acquiescence—which a simple longstring or low-variability rule already catches perfectly. Third, even at its best a single threshold leaves a gap to the oracle, because no label-free rule can perfectly separate the two classes.

Each limit is the mirror image of a strength of some other detector. Mahalanobis distance and longstring catch the consistent patterns ReReReRe misses; low-variability indices catch straightlining outright. This complementarity suggests that the right way to use ReReReRe is not alone but as one informative feature among several, combined by a classifier that learns when to trust each. That is the subject of Chapter 4, which embeds ReReReRe in a Random Forest ensemble alongside the auxiliary detectors and quantifies, with the confidence intervals introduced in Section 2.7, exactly how much the coherence signal contributes.

References

- Curran, P. G. (2016). Methods for the detection of carelessly invalid responses in survey data. *Journal of Experimental Social Psychology*, 66, 4–19. <https://doi.org/10.1016/j.jesp.2015.07.006>
- Meade, A. W., & Craig, S. B. (2012). Identifying careless responses in survey data. *Psychological Methods*, 17(3), 437–455. <https://doi.org/10.1037/a0028085>